

DSA0607 DATA HANDLING AND VISUALIZATION

CO3_AT3_ INTREPRETATION

1. An e-commerce company is analyzing the purchase amounts of 10,000 customers during a festive sale. Most customers spent between ₹500 and ₹2,000, while a small number of customers made purchases above ₹25,000. The company wants to understand customer spending behavior, identify high-value customers, and determine whether the spending distribution follows a normal pattern. How would you use ECDF, histograms, and Q-Q plots to analyze the spending data, and what insights could be derived from the results?

Aim

To analyze customer purchase amounts using **Histogram, ECDF, and Q-Q Plot** in Tableau to understand spending distribution, identify high-value customers, and assess whether the spending data follows a normal distribution.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Connect to **Customer_Spending.csv**.
3. Verify the fields and open **Sheet 1**.

2. Create a Histogram

1. Right-click **Purchase_Amount** → **Create** → **Bins**.
2. Drag **Purchase_Amount (bin)** to **Columns**.
3. Drag **Customer_Spending.csv (Count)** to **Rows**.
4. Rename the worksheet as **Histogram**.

3. Create an ECDF

1. Open a **New Worksheet**.
2. Drag **Purchase_Amount** to **Columns**.
3. Drag **Customer_Spending.csv (Count)** to **Rows**.
4. Right-click the measure → **Quick Table Calculation** → **Running Total**.

5. Change the **Marks** type to **Line**.
6. Rename the worksheet as **ECDF**.

4. Create a Q-Q Plot

1. Import a dataset containing **Theoretical_Quantile** and **Sample_Quantile**.
2. Drag **Theoretical_Quantile** to **Columns**.
3. Drag **Sample_Quantile** to **Rows**.
4. Change **Marks** to **Circle**.
5. Add a **Trend Line** from the **Analytics** pane.
6. Rename the worksheet as **Q-Q Plot**.

5. Create the Dashboard

1. Click **New Dashboard**.
2. Add the **Histogram**, **ECDF**, and **Q-Q Plot** worksheets.
3. Add a filter for **Customer_ID** (optional).
4. Save the dashboard.

Explanation

Histogram

A histogram shows the frequency distribution of customer purchase amounts. It helps identify the most common spending range and highlights customers with unusually high purchase values.

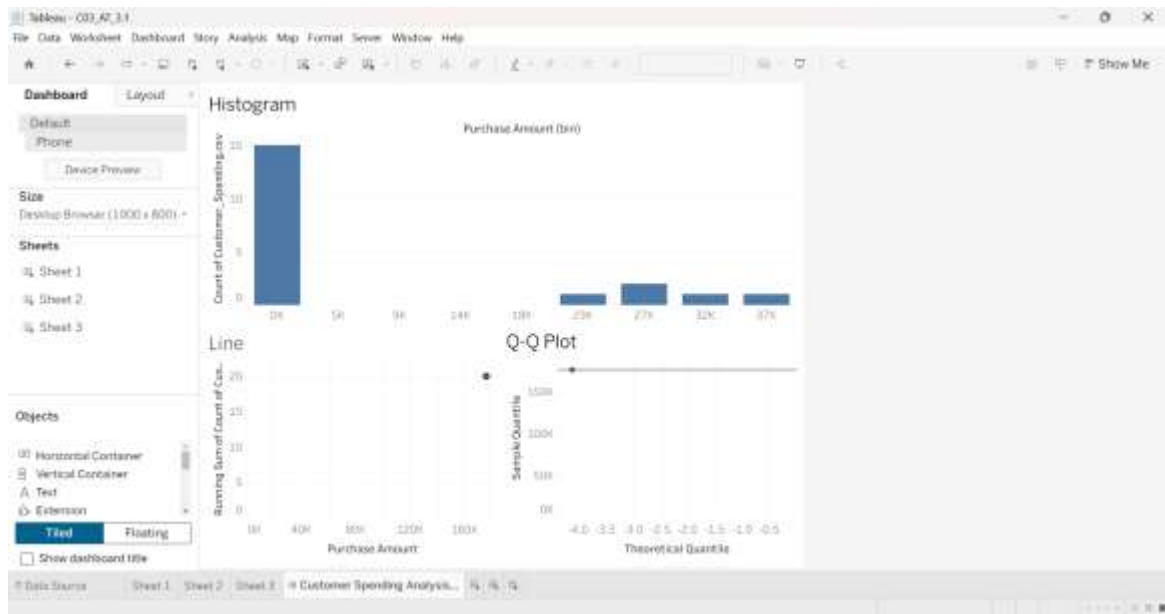
ECDF (Empirical Cumulative Distribution Function)

An ECDF displays the cumulative percentage of customers whose purchases are less than or equal to a given amount. It helps compare spending patterns and determine the proportion of customers within different purchase ranges.

Q-Q Plot

A Q-Q plot compares the purchase amount distribution with a theoretical normal distribution. Points close to the reference line indicate normality, while deviations from the line suggest skewness or the presence of outliers.

Output



Result

The Tableau dashboard successfully visualizes customer spending using **Histogram**, **ECDF**, and **Q-Q Plot**. It helps identify high-value customers, detect outliers, evaluate spending distribution, and determine whether purchase amounts follow a normal distribution.

2.A hospital is comparing patient waiting times in two departments: Emergency Care and General Consultation. The management wants to determine whether waiting times in both departments follow similar distributions and whether one department experiences significantly longer delays. Explain how Q-Q plots, boxplots, and density plots can be used to compare the two datasets and evaluate distributional differences.

Aim

To compare patient waiting times in the **Emergency Care** and **General Consultation** departments using **Q-Q Plots**, **Box Plots**, and **Density Plots** in Tableau to evaluate distributional differences and identify delays.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.

2. Connect to **Hospital_Waiting_Time.csv**.
3. Verify the fields and open **Sheet 1**.

2. Create a Box Plot

1. Drag **Department** to the **Columns** shelf.
2. Drag **Waiting_Time** to the **Rows** shelf.
3. Click **Show Me** → **Box-and-Whisker Plot**.
4. Rename the worksheet as **Box Plot**.

3. Create a Density Plot

1. Open a **New Worksheet**.
2. Drag **Waiting_Time** to **Columns**.
3. Drag **Department** to **Color**.
4. Select **Show Me** → **Density Plot** (or use the Density mark if available).
5. Rename the worksheet as **Density Plot**.

4. Create a Q-Q Plot

1. Import a dataset containing **Theoretical_Quantile** and **Sample_Quantile**.
2. Drag **Theoretical_Quantile** to **Columns**.
3. Drag **Sample_Quantile** to **Rows**.
4. Change **Marks** to **Circle**.
5. Add a **Trend Line** from the **Analytics** pane.
6. Rename the worksheet as **Q-Q Plot**.

5. Create the Dashboard

1. Click **New Dashboard**.
2. Add the **Box Plot**, **Density Plot**, and **Q-Q Plot** worksheets.
3. Add a **Department** filter (optional).
4. Save the dashboard.

Explanation

Box Plot

A box plot compares the waiting-time distributions of Emergency Care and General Consultation. It shows the median, quartiles, variability, and outliers, helping identify which department has longer waiting times.

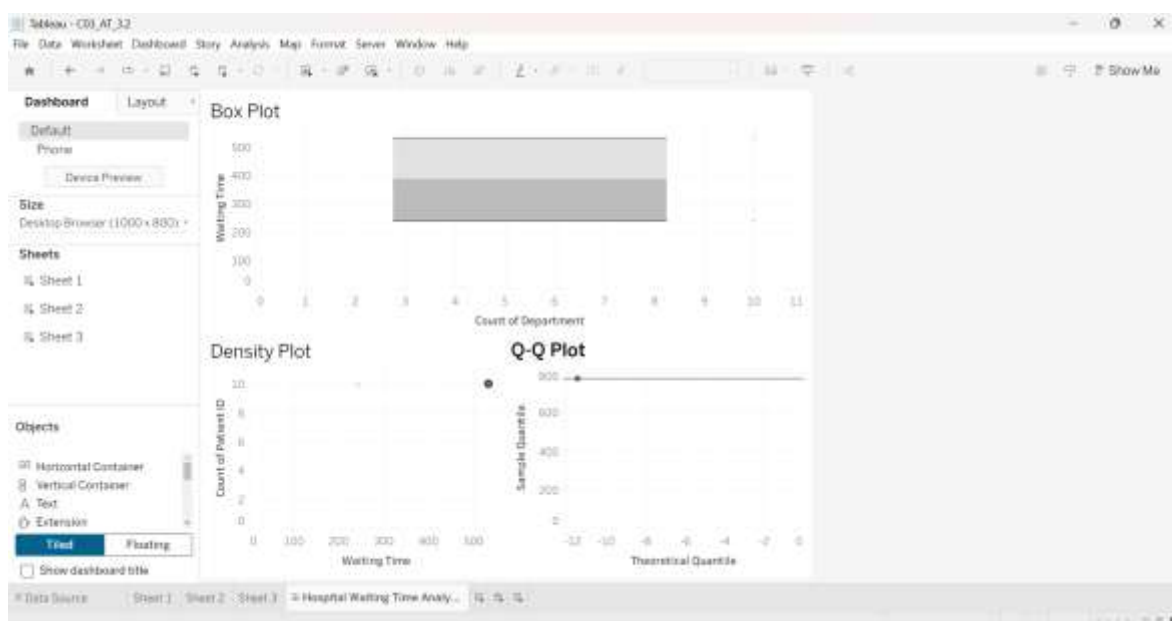
Density Plot

A density plot displays the distribution of waiting times for each department. Wider peaks indicate where waiting times are concentrated, making it easier to compare overall distribution patterns.

Q-Q Plot

A Q-Q plot compares the observed waiting-time distribution with a theoretical normal distribution. If the points closely follow the reference line, the waiting times are approximately normally distributed. Large deviations indicate skewness or outliers.

Output



Result

The Tableau dashboard successfully compares patient waiting times using **Box Plots**, **Density Plots**, and **Q-Q Plots**. It helps identify differences in waiting-time distributions, detect outliers, evaluate delays between departments, and assess whether the data follows a normal distribution.

3. A digital marketing team wants to present the number of likes received by five social media campaigns: 2,000, 5,000, 8,000, 15,000, and 30,000 likes. One analyst suggests using a 3D exploded pie chart, while another recommends a horizontal bar chart. Discuss how each visualization may influence audience perception and identify the most effective method for accurately comparing campaign

performance. What design principles should be followed to avoid misleading interpretations?

Aim

To compare the performance of five social media campaigns using appropriate visualizations and identify the most effective chart for accurate comparison while following good visualization design principles.

Procedure

1. Import the Dataset

1. Open **Tableau Desktop/Public**.
2. Connect to **Campaign_Likes.csv**.
3. Verify the fields and open **Sheet 1**.

2. Create a Horizontal Bar Chart

1. Drag **Campaign** to the **Rows** shelf.
2. Drag **Likes** to the **Columns** shelf.
3. Select **Bar** in the **Marks** card.
4. Sort the bars in descending order of likes.
5. Rename the worksheet as **Horizontal Bar Chart**.

3. Create a Pie Chart

1. Open a **New Worksheet**.
2. Drag **Campaign** to **Color**.
3. Drag **Likes** to **Angle**.
4. Change the **Marks** type to **Pie**.
5. Rename the worksheet as **Pie Chart**.

4. Create the Dashboard

1. Click **New Dashboard**.
2. Add the **Horizontal Bar Chart** and **Pie Chart**.
3. Add a dashboard title: "**Social Media Campaign Performance Dashboard**".
4. Save the dashboard.

Explanation

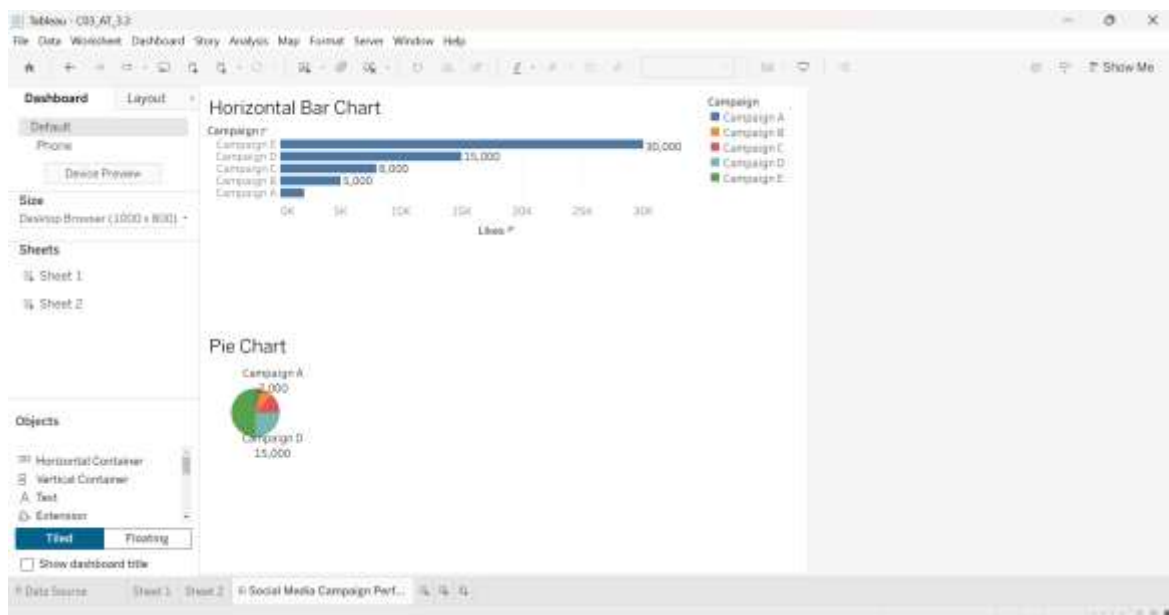
Horizontal Bar Chart

A horizontal bar chart accurately compares the number of likes across campaigns using bar lengths. It makes it easy to identify the highest- and lowest-performing campaigns and supports precise comparisons.

Pie Chart

A pie chart shows the proportion of total likes contributed by each campaign. However, when there are large differences between values, comparing slice sizes becomes difficult. A 3D exploded pie chart can further distort visual perception and may exaggerate some slices.

Output



Result

The **Horizontal Bar Chart** is the most effective visualization for comparing campaign performance because it enables accurate comparison of likes across campaigns. Although a pie chart can show each campaign's share of total likes, a **3D exploded pie chart is misleading** and should be avoided. Following good visualization principles ensures clear, accurate, and reliable communication of campaign performance.